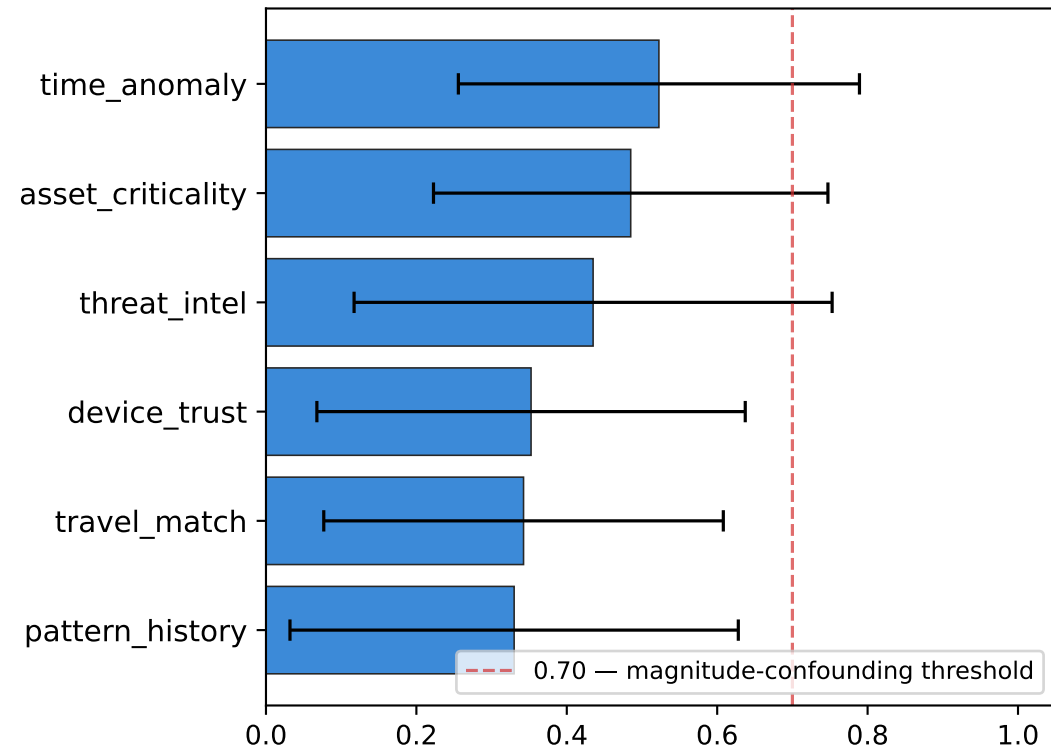


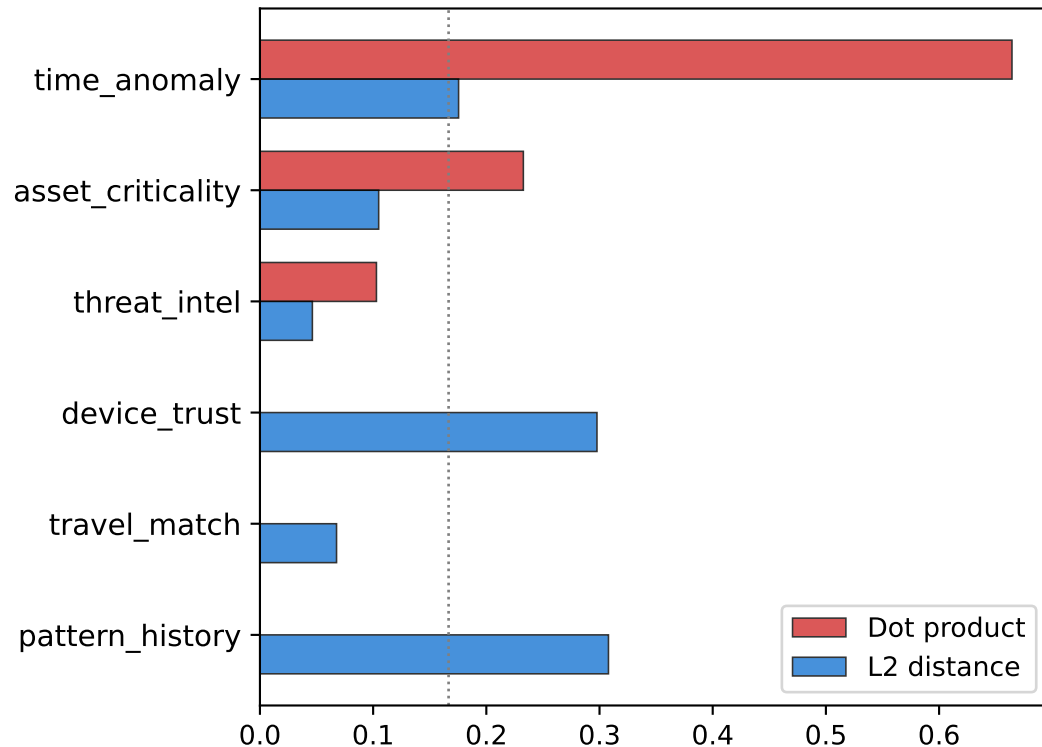
EXP-C1: Why Dot Product Fails on Bounded [0,1] Features — Magnitude Confounding

Panel A: Mean centroid value per factor
(averaged over all category × action pairs)



Red bars: factors with mean < 0.70 (all have near-constant dot product scores regardless of discriminative value).

Panel B: Discriminative contribution per factor
(separating power: action 0 vs action 1, category 0)



Fraction of total separating power (normalized)

Dot product
L2 distance

Concrete example (device_trust):

Dot product: $0.90 \times 0.88 = 0.79$ (constant for all actions)

L2 distance: $(0.90 - 0.88)^2 = 0.0004$ (near-zero — not discriminative)

→ dot product weight is driven by magnitude, not by deviation